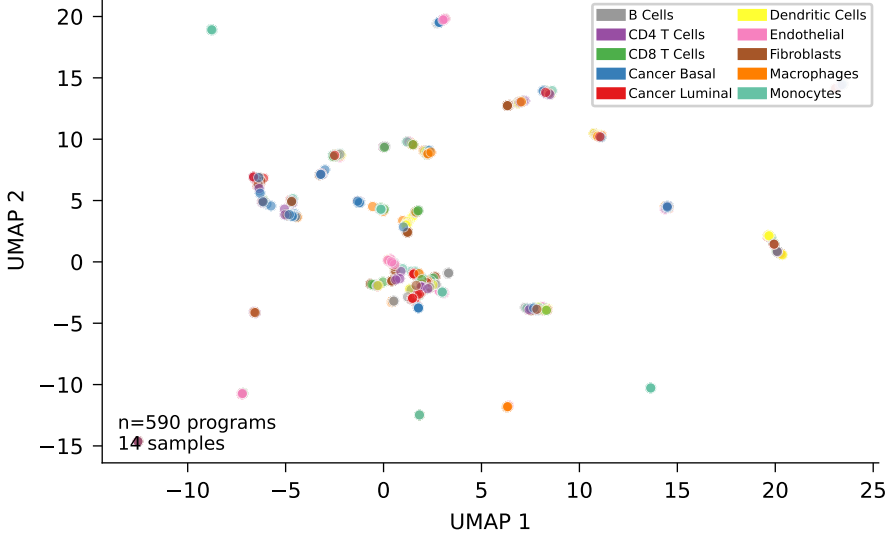


A. Module 5: Cross-Sample Integration

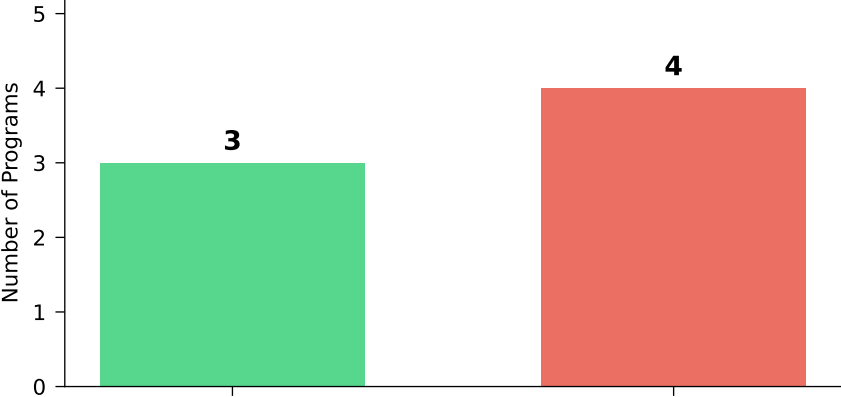
Panel A: Cross-Sample Integration Schematic
(Create in vector editor)

- Show:
- 14 patient samples (P1-S1, P1-S2, ...)
 - Per-sample program discovery (Module 4)
 - Harmony alignment in latent space
 - Unified program identities across samples

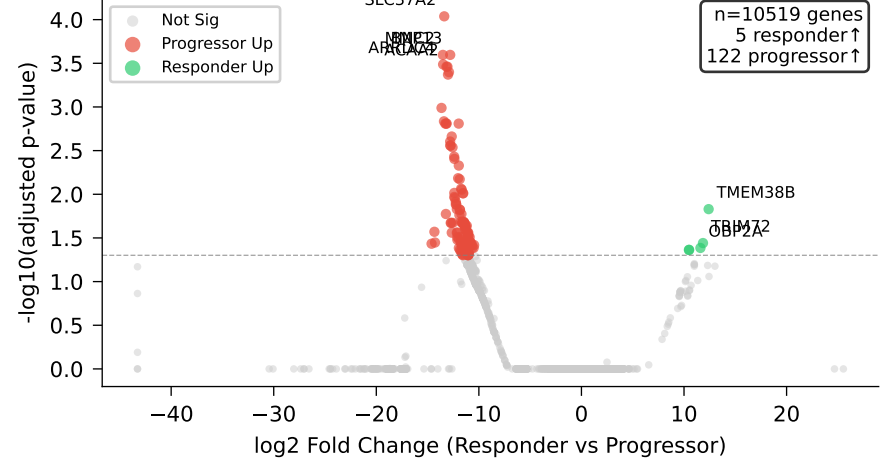
B. Program Embedding (colored by cell type)



C. Response-Associated Programs



D. Differential Expression (Pseudo-bulk PyDESeq2)



E. Conserved Spatial Relationships

Responder-enriched
Macrophages: FABP4, HBA2, TNXB
Macrophages: VIM, TMSB4X, S100A6
Cancer_Luminal: MGP, MT-CO3, FOS

Progressor-enriched
Dendritic_Cells: FTL, FN1, TIMP1
Cancer_Basal: MT-CO3, MT-ND4, MT-CO2
Cancer_Luminal: MGP, HBA2, LTF

